

Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026
Homework #8 Solutions

1. (a) $E(Z) = E(X) + E(Y) = \frac{5}{2} + \frac{1}{4} = \frac{11}{4} = 2.75$.
- (b) Since X and Y are independent, $\text{Var}(Z) = \text{Var}(X) + \text{Var}(Y)$. We can take these variances from our table of common random variables to get $\text{Var}(Z) = \frac{25}{12} + \frac{1}{16} = \frac{103}{48} = 2.15$.
- (c) To get the full PDF of Z , we need to do what's called the convolution integral of the PDFs of X and Y . If you took Signals, this will be familiar; otherwise doing this is a preview of what to expect! There will be three cases. When $z < 0$, the PDF is 0. We can solve the other cases using the exponential CDF without doing a lot of integration. When $z \in [0, 5]$, we have partial overlap between the PDFs:

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{\infty} f_Y(x) f_X(z-x) dx \\ &= \int_0^z \frac{1}{5} \cdot 4e^{-4x} dx \\ &= \frac{1}{5}(1 - e^{-4z}) \end{aligned}$$

When $z > 5$, we have full overlap between the PDFs:

$$\begin{aligned} f_Z(z) &= \int_{-\infty}^{\infty} f_Y(x) f_X(z-x) dx \\ &= \int_{z-5}^z \frac{1}{5} \cdot 4e^{-4x} dx \\ &= \frac{1}{5}e^{-4z}(e^{20} - 1) \end{aligned}$$

So the overall PDF is:

$$f_Z(z) = \begin{cases} 0 & z < 0 \\ \frac{1}{5}(1 - e^{-4z}) & z \in [0, 5] \\ \frac{1}{5}e^{-4z}(e^{20} - 1) & z > 5 \end{cases}$$

2. (a) First we need the mean and variance of a Smiski's weight:

$$\begin{aligned} E(X) &= \int_0^2 x \cdot \frac{3}{2}x - \frac{3}{4}x^2 dx \\ &= \left. \frac{1}{2}x^3 - \frac{3}{16}x^4 \right]_{x=2}^{x=0} \\ &= 1 \end{aligned}$$

$$\begin{aligned} \text{Var}(X) &= E(X^2) - (E(X))^2 \\ &= \int_0^2 x^2 \cdot \frac{3}{2}x - \frac{3}{4}x^2 dx - 1 \\ &= \left. \frac{3}{8}x^4 - \frac{3}{20}x^5 \right]_{x=2}^{x=0} - 1 \\ &= \frac{1}{5} \end{aligned}$$

Now if we let $S_{100} = \sum_{i=1}^{100} X_i$ and $M_{100} = \frac{1}{100} \sum_{i=1}^{100} X_i$, we want to estimate

$$\begin{aligned} P(S_{100} \in [85, 115]) &= P(|S_{100} - 100| \leq 15) \\ &= P(|M_{100} - 1| \leq \frac{15}{100}) \\ &\geq 1 - \frac{1}{100 \cdot \left(\frac{15}{100}\right)^2} \frac{1}{5} \\ &= 1 - \frac{20}{225} \\ &= \frac{205}{225} \\ &= 0.91 \end{aligned}$$

Here we're using the key Chebyshev bound that

$$P(|M_n - \mu| < \epsilon) \geq 1 - \frac{1}{n\epsilon^2} \sigma^2$$

- (b) The Central Limit Theorem says that

$$P(|S_n - n\mu| \leq z\sigma\sqrt{n}) = 1 - 2Q(z)$$

Here, $\mu = 1$, $\sigma^2 = \frac{1}{5}$, $n = 100$ and $z\sigma\sqrt{n} = 15$. That is, $z = \frac{15}{10} \sqrt{5} = 3.35$. So

$$P(|S_{100} - 100| \leq 15) = 1 - 2Q(3.35) = 1 - 2(0.000404) = 0.999192$$

- (c) We can see the actual probability is much higher than we estimated with Chebyshev. The Central Limit Theorem is much more accurate when n is large since it does a good job of actually reflecting the PDF of S_n .

3. (a) X is uniform on $[4, 6]$, so $\mu = 5$ and $\sigma^2 = \frac{1}{3}$. The Central Limit Theorem says

$$P\left(\frac{S_n - n\mu}{\sigma\sqrt{n}} \geq z\right) \approx Q(z)$$

or

$$P(S_n \geq n\mu + z\sigma\sqrt{n}) \approx Q(z)$$

Here $n = 35$ so

$$35 \cdot 5 + z \left(\sqrt{\frac{35}{3}} \right) = 180$$

Thus

$$z = \frac{180 - 175}{\sqrt{\frac{35}{3}}} = 1.46$$

So the probability is about $Q(1.46) = 0.072$.

- (b) X is Poisson with $\mu = \alpha = 4$, and hence $\sigma^2 = \alpha = 4$ also. We require that

$$P\left(\frac{S_{10} - 10 \cdot 4}{\sqrt{4}\sqrt{10}} \geq z\right) \leq 0.02$$

That is, we must find z so that $Q(z) = 0.02$. Wolfram Alpha says $z = 2.05$. Therefore

$$(10)(4) + (2.05)(2\sqrt{10}) = 52.97$$

Thus NASA should provide Reid with 53 SD cards on the mission.

- (c) The confidence interval for the mean is

$$M_{31} \pm z_{0.025} \frac{\sigma}{\sqrt{31}}$$

From Table 8.1, $z_{0.025} = 1.96$. We are told $\sigma = 20$, and our sample mean $M_{31} = \frac{2026}{31} = 65.355$. So the 95% confidence interval is

$$65.355 \pm \frac{(1.96)20}{\sqrt{31}} = 65.355 \pm 7.041 = [58.314, 72.396]$$